

Team Worksheet — Lab 13

Nested and Destructive Gauge Studies (Black Belt Only)

Team members: _____

Date: _____

1. Our understanding of the problem

2. Our working

Use the data in `lab-13-nested-destructive-gauge.xlsx`.

3. What we produced

Deliverable: A nested gauge study design and analysis for a destructive or single-measurement test.

4. What the evidence does NOT tell us

5. Our conclusion

6. Step notes

Step 1. State precisely why a crossed design is invalid for destructive tests: the part no longer exists to be measured a second time.

Step 2. Identify a destructive or non-repeatable measurement in your capstone — or use the Meridian seal-strength test on infusion pump tubing.

Step 3. Understand the nested design: each operator measures DIFFERENT parts, and parts are nested within batch rather than crossed with operator.

Step 4. Establish homogeneous batches. The design assumes parts within a batch are effectively identical — if they are not, part variation contaminates repeatability.

Step 5. Justify batch homogeneity: same lot, same machine, same setup, consecutive production, minimal elapsed time.

Step 6. Design the study: 3 operators, 10 batches, 3 parts per operator per batch, all parts within a batch treated as replicates.

Step 7. Run the study and record results in the nested layout — operator within batch, not operator crossed with part.

Step 8. Compute variance components for repeatability and reproducibility using the nested model. Note that repeatability is now confounded with within-batch part variation.

Step 9. State that confounding explicitly in your report. A nested study OVERSTATES repeatability variation, so it is a conservative test.

Step 10. Decide accept/reject, and record the design choice and its rationale for the Day 5 presentation — expect this to be challenged.

7. Self-check

You can explain why a crossed study is invalid for your measurement, your batches are justified as homogeneous, and your report states the repeatability/part-variation confounding explicitly.

Complete? ■ Yes ■ Not yet

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